MAE 1001

Introduction to Mechanical and Aerospace Engineering

Course website: https://gwu-mae1001.github.io/fall2025/

Prof. Kartik Bulusu, MAE Dept.

Graduate Teaching Assistant:
Preethi Siva Kumar (MAE Dept., MS student)

Learning Assistants:
Nathan Jannsen (Senior, MAE Dept.)
Olivia Falciani (Senior, MAE Dept.)
Shota Kakiuichi (Senior, MAE Dept.)

Tech support:
Rithik Gunuganti (CS Dept., MS student)

School of Engineering & Applied Science
THE GEORGE WASHINGTON UNIVERSITY

Fall 2025

Photo: Kartik Bulusu

Seating Arrangement

Table 1 (8 students):
- Curtis Charlery
- Daris Dedic
- Johnny Lopez Bertrand
- Luz De La Cruz
- Ryan Barks
- Teresa Cherian
- Tre Troy
- Jessica Gil

Table 2 (8 students):
- Irene Oh
- Henry Steffen
- Lucas Narvaez
- Novah Wali
- Zaynah Asfour
- Evan Chukwunonye
- Cadence Cassidy
- Owen Fritts

Table 3 (8 students):
- Diego Flores
- Nate Behabtu
- Omar Mohammed A I Al Emadi
- Alex Lagrant
- Addison Dooley
- Holt Perryman
- Cesar Tovar
- Lilia Hammadi

Table 4 (8 students):
- Grace Ephrem
- Zachary Bogan
- Aidan Willett
- Dennis Ventura
- Matheo Valdivia
- Yahya Ahmed Ahmed Sadek Elsewedy
- Sena Mute
- Otto Erhart

Table 5 (8 students):
- Wassime Ait-Si
- Naomi Kim
- Tim Harrington
- Gael Merida Chavez
- Camryn Lorencovitz
- Emre Ozkazanc
- Alexander Lele
- Albert Wu

Table 6 (7 students):
- Maeve Carey
- Ivan De La Cruz
- Thomas Melotto
- Harper Barrios
- Shu Shu Wu
- Nabek Ababiya
- Andy Pham

Table 7 (7 students):
- Ali Ahmed Y A Al Malki
- Paige Romanowski
- Evan Agapiou
- Saihaj Kaur
- Brandon Alvarado
- Dylan Nichols
- Ethan Peralta

Table 8 (7 students):
- Jocelyn Carrillo
- Marthine Jean Pierre
- Daniel Zhang
- Inti Stephenson
- Akina Pallera
- Stella Iyengar
- Eli Ecret

Table 9 (7 students):
- Chance Hwang
- Kazi Afra Saiara
- Mohammed Jabir H A Al Jabir
- Cody Keel
- Kim Zamora
- Malik Ali
- Hamza Chaudry

Road Map for Class 5

30-Minute Session Blocks Breakdown

Photo Credits: Kartik Bulusu
Icon source: canva.com

Introduction to Week 5

Announcements, overviews and more!

Blitz Talk with Dominic Savarino

Guest Lecture with Dr Peng Wei

Energy conversion technologies involving advanced materials

Screwdriver Disassembly Lab

Dissect a cordless screwdriver

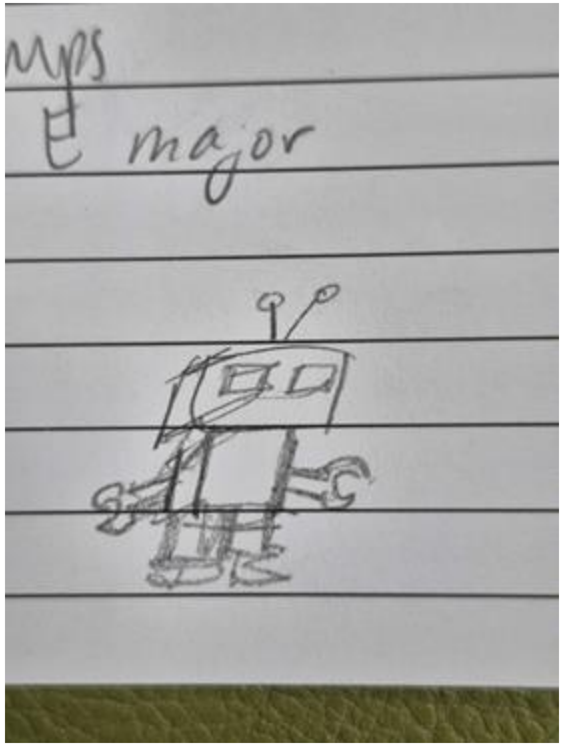
LaTex 101

Master LaTeX and Overleaf with ease!

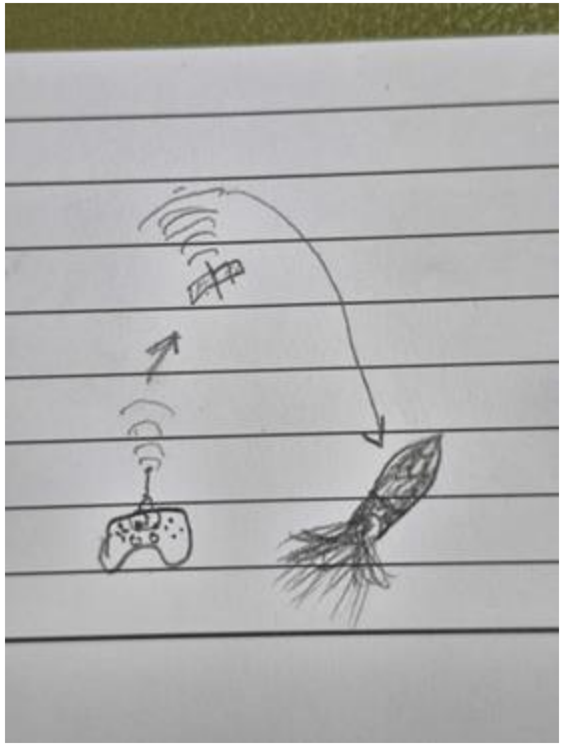
Raspberry Pi

Achieve the highest score in Snake.

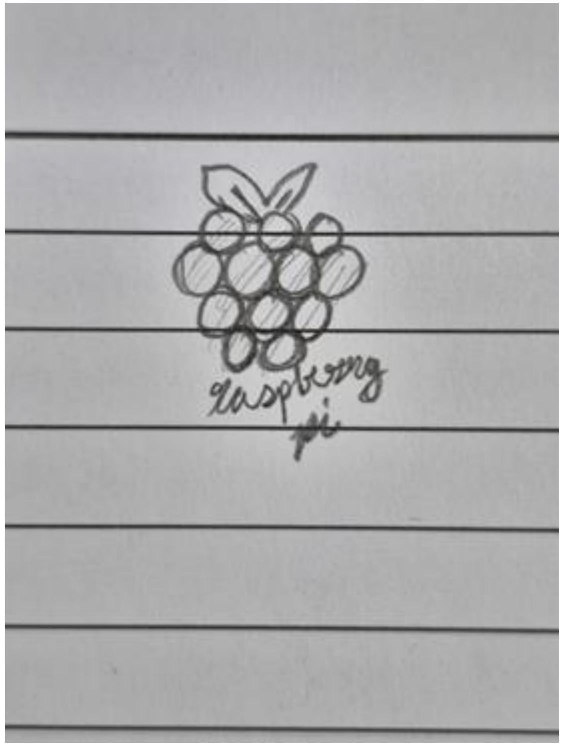
Week 4 - Best Doodles



Shu Shu Wu

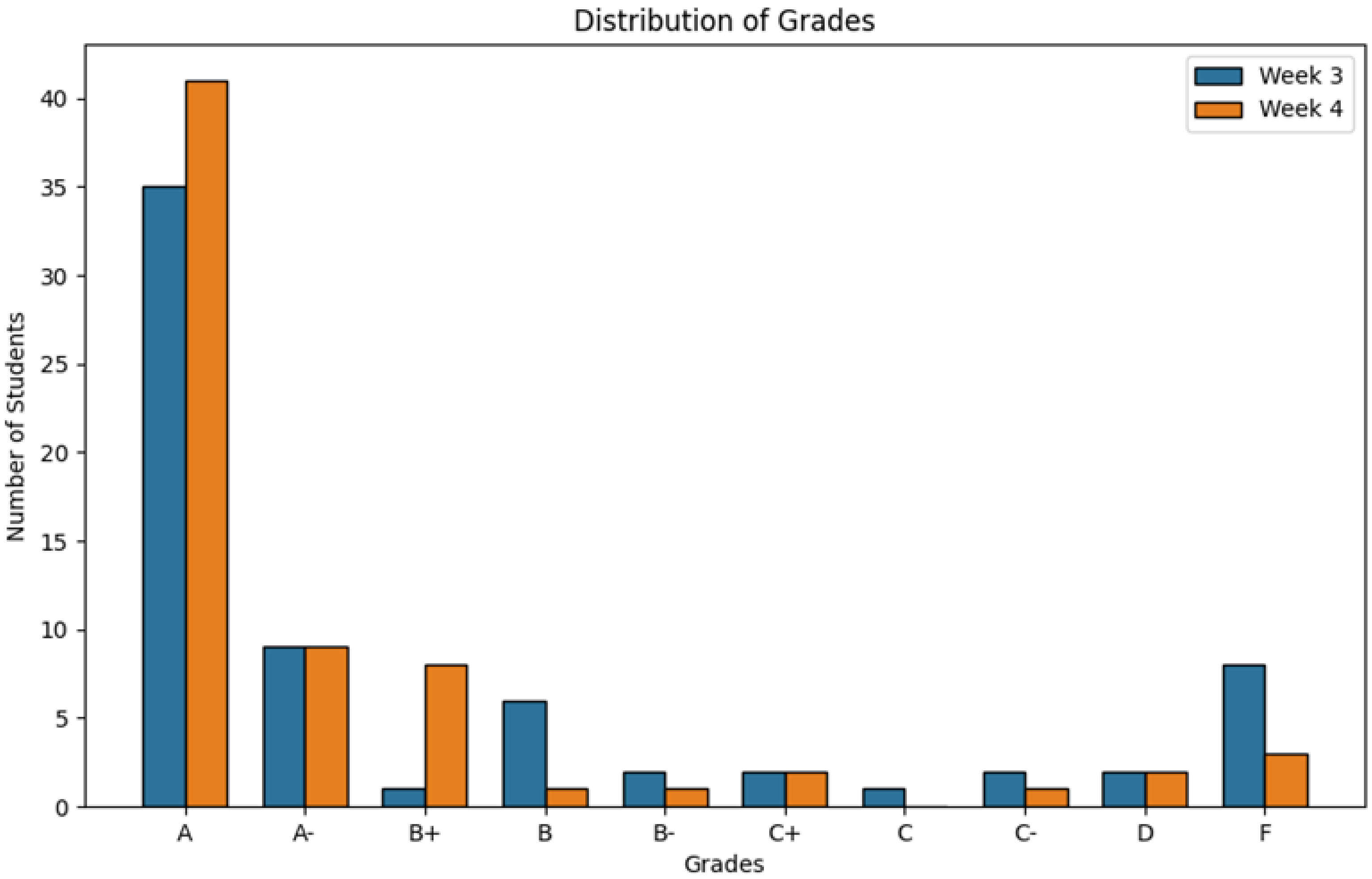


Matheo Valdivia



Grace Ephrem

Week 4 - Grade Distribution



Score	Grade
90-100	A
84-89	A-
78-83	B+
72-77	B
66-71	B-
60-65	C+
56-59	C
52-55	C-
46-51	D
0-45	F

Blitz Talk with Dominic Savarino

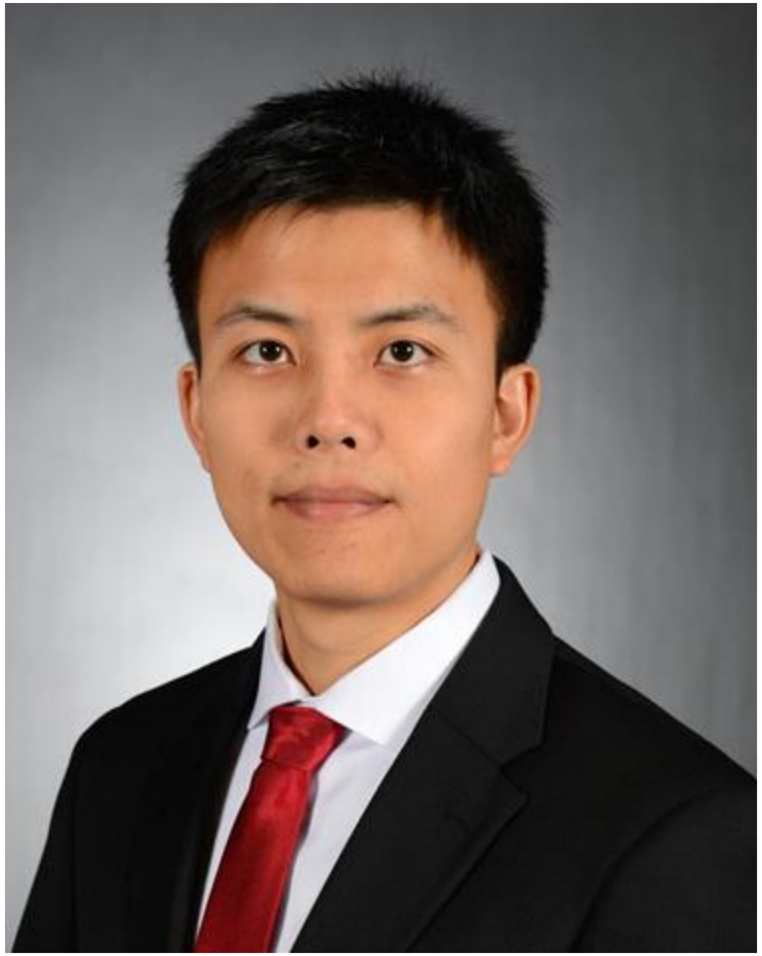
Senior in Mechanical and Aerospace Engineering

Email: dsavarino@email.gwu.edu

LinkedIn: linkedin.com/in/dominic-savarino/

Dominic Savarino (BS ’25, MS ’26) is a MAE teaching assistant and graduate student studying robotics and controls. While an undergraduate at GW, he completed internships with PennDOT, Space Telescope Science Institute, and GW Intelligent Aerospace Systems Lab, studied abroad, and was a part of organizations such as Clark Scholars, SEASSPAN, ASME, and Pi Tau Sigma.

Guest Lecture with Professor Peng Wei



Associate Professor of Mechanical and Aerospace Engineering

Email: pwei@gwu.edu

Website: web.seas.gwu.edu/pwei/

Peng Wei is an associate professor in the Department of Mechanical and Aerospace Engineering at the George Washington University, with courtesy appointments at Electrical and Computer Engineering Department and Computer Science Department. By contributing to the intersection of control, optimization, machine learning, and artificial intelligence, his research group develop autonomous functions and decision support tools for aviation, avionics and aerial robotic systems. His current focuses are (1) safety, efficiency, and scalability of aircraft autonomy, multi-agent autonomy and human-autonomy teaming; (2) aviation applications including air traffic management/control (ATM/C), airline operations, UAS traffic management (UTM), advanced air mobility (AAM), and aviation electrification; and (3) AI safety, security and certification for safety-critical systems. He is also leading the Intelligent Aerospace Systems Lab (IASL) and is an AIAA Associate Fellow.

Exit Ticket

Name:

Three things you learnt today:

1.

2.

3.

Best Part of the Class:

Most Challenging Part of the Class:

Assignments for Week 5 - Due 12/04

In-Class Lab 5A: Bias in Dice Experiment - Trial 4

In-Class Lab 5B: Screwdriver Disassembly - Lab Notebook

In-Class Lab 5C: RPi Sense Hat Snake Game Score

Assignment 5D: Overleaf Report Submission